Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Proposal

Proponent/Researcher:	**CW10**
Project Title: Tracking and Reaching ALS Learners System with Geo-mapping and Decision-Support System for Santa Cruz, Laguna	
Introduction	Mapping extends beyond simply pinpointing locations; it involves understanding the movement and needs of people within a community. In the context of education, effective mapping enables stakeholders to identify gaps in access and learning, make informed decisions on resource allocation, and design targeted interventions that address the specific needs of learners. The Alternative Learning System (ALS) in Sta. Cruz, Laguna, plays a critical role in providing accessible and flexible education to out-of-school youth, adult learners, and marginalized populations. Despite progress in enrollment and completion rates, there remain significant challenges in tracking and supporting both ALS graduates and individuals who have yet to enroll in the program. To overcome these challenges, the Geo Mapping System and Decision Support System for Tracking ALS Graduates and At-Risk Learners integrates advanced geospatial artificial intelligence with mobile technology. This innovative system empowers ALS coordinators, educators, and local government units to accurately map and monitor the distribution of ALS learners across Sta. Cruz. Complemented by a robust Decision Support System (DSS), the platform analyzes data to identify priority areas, forecast enrollment trends, and generate actionable insights for more strategic educational planning. Furthermore, the system includes a learner-centric performance monitoring portal designed to enhance engagement and support self-paced learning. Through this portal, learners can access educational materials, complete assessments, track their progress, receive announcements, and communicate with facilitators. By combining spatial visualization with intelligent data analysis, this comprehensive approach aims to improve outreach effectiveness, optimize resource distribution, and ultimately ensure that every learner in the community receives the support they need for success.
Statement of the Problem	Despite the ongoing efforts of the Alternative Learning System (ALS) in Sta. Cruz, Laguna to connect with out-of-school youth and adult learners, there's still a major hurdle in accurately identifying and locating ALS graduates and at-risk learners, those who are not currently enrolled or are at risk of dropping out. This lack of precise geographic and status-based information makes it difficult for ALS coordinators and local authorities to effectively reach out, allocate resources wisely, and provide timely support to the learners who need it most. Without a solid mapping and data-driven support system, educational services struggle to meet learners where they are. This results in lower enrollment, reduced program

	completion, and missed opportunities for re-engagement. The absence of real-time, location-based insights also makes it challenging to implement targeted interventions that address the unique needs of each community. Moreover, without a centralized platform for learners to monitor their own academic progress, many at-risk individuals lack the tools to stay motivated and connected. To address this, the proposed system integrates a performance monitoring portal where learners can access educational content, track module completion, and communicate directly with facilitators. This dual approach, combining geospatial outreach tools with learner performance tracking ensures that both educators and students have the insights they need to succeed.
Objective of the Study	To develop a Geo-mapping and decision support system that accurately tracks the geographic locations of ALS finishers and non-ALS students in Sta. Cruz, Laguna to enhance educational outreach, data-driven planning, and support services. **Specific Objectives:** 1. To create an interactive, mobile-accessible map that displays the real-time locations of ALS finishers and students not yet enrolled in ALS programs. 2. To utilize geo-mapping techniques to analyze spatial patterns and identify areas with low ALS participation or high dropout rates. 3. To develop a Decision Support System (DSS) that generates actionable insights and recommendations for ALS coordinators and local government units, based on spatial and demographic data. 4. To enable data collection and updating of student information through a user-friendly mobile application accessible by field workers and educators. 5. To implement a student performance monitoring portal that provides learners with access to downloadable modules, quizzes, grades, feedback, certification progress, and messaging features for enhanced engagement and self-tracking 6. To improve the overall enrollment and completion rates of ALS programs in Sta. Cruz, Laguna by addressing geographic barriers and enhancing strategic decision-making.
Scope and Limitation of the Study	The goal of this project is to develop an effective Geo-mapping and Decision Support System (DSS) that enables accurate identification and visualization of the geographic locations of ALS finishers and non-ALS students in Sta. Cruz, Laguna. This system aims to improve the planning, outreach, and delivery of ALS services by providing educators and local authorities with real-time, actionable spatial data and analytical insights. **Scope:** ● The system will focus on tracking and mapping the geographic locations of two key groups: ALS graduates and learners who are not yet enrolled in the ALS program within Sta. Cruz, Laguna. ● It will utilize Geo-mapping technologies in combination with mobile applications to collect, analyze, and visualize spatial data related to learner distribution and outreach needs.

	A mobile application will allow field workers and ALS coordinators to input and update learner data, including geographic location, enrollment status, and basic learner profiles. ● A built-in Decision Support System (DSS) will analyze collected data and generate automated reports and strategic recommendations to support evidence-based decisions on outreach, resource allocation, and intervention planning. ● The system will also include a performance monitoring portal for learners, providing access to downloadable learning materials, topic-based modules, quizzes, grades, facilitator feedback, and certificate progress tracking. It will also feature announcements, messaging functions, and offline access for areas with limited internet connectivity. ● The project will primarily cover data collection and system deployment within Sta. Cruz, Laguna, with support for offline data entry and periodic synchronization when internet access is available. **Limitations:** ● The accuracy of geographic data depends on the availability and quality of GPS or mobile network signals, which may be inconsistent in remote or densely populated areas. ● The effectiveness of the system relies on the active participation and cooperation of students, families, ALS personnel, and field workers to provide timely and accurate data. ● Privacy and data protection concerns may limit the granularity of data collected or displayed, especially in relation to individual learner identities and locations. ● While the system includes a learning portal and tracking features, it will not directly address curriculum content delivery or deeper factors influencing learner motivation. ● Implementation, training, and maintenance of the system require technical knowledge and sustainable support, which may pose a challenge for long-term deployment if resources are limited.
	Theme 1: Geographic Information System The rapid growth of Geographic Information Systems (GIS) over the past decade has significantly expanded the understanding of GIS technology. Originally viewed primarily as a collection of digital entities, models, and tools used to generate or manipulate geographical information, GIS has evolved into a mediating instrument through which humans experience, navigate, and interpret the world. To better capture this expanded scope, the humanistic GIS perspective has been introduced. This approach, deeply grounded in humanistic geography, offers a systematic framework that integrates the currently fragmented body of humanism-related GIS research. It also repositions the epistemological foundation of GIS by emphasizing its role in mediating human experience rather than solely functioning as a technical system (Zhao, 2022). This paper analyzes the benefits of Geographic Information Systems in urban planning and the tourism sector through a systematic literature review. It discusses how spatial data is crucial in maximizing the effectiveness of land use and urban planning applications. The study highlights the implementation of machine learning models in geospatial datasets, which can be adapted for educational planning and outreach programs (C. Fauzi, 2024).

Review of Literatures and Related Systems

This paper reviews the integration of Geographic Information Systems in cartography, summarizing current research trends in using cartographic design. It identifies various GIS methods and their applications in cartographic tasks such as generalization, symbolization, and map interpretation. The study also addresses ethical considerations, which are crucial when implementing Geo-mapping in educational settings (Kang et al., 2023).

Theme 2: Alternative Learning Systems (ALS) and Non-Formal Education in the Philippines

This study examines two ALS programs—Don Bosco PUGAD and iCare YESNova—focusing on how they foster agency among marginalized out-of-school youth (OSY). Utilizing Amartya Sen's Capabilities Approach and Bourdieu's Social Reproduction Theory, the research highlights both the empowering aspects and the structural limitations of ALS in addressing educational disparities (Reyes, n.d.)

This qualitative study explores the academic and non-academic experiences of non-formal education graduates enrolled in a teacher education program in the Philippines. It identifies their unique needs and proposes a program framework to support their success in higher education (Saquing, 2023).

This literature review examines the teaching and learning of mathematics within the ALS framework. It highlights the challenges faced by learners and educators, including curriculum design, teaching strategies, and assessment methods. The study suggests improvements to enhance mathematical learning in non-formal education settings (Velasco & Lomibao, 2024).

Theme 3: Data-Driven Educational Outreach and Intervention Strategies

This study evaluated the effectiveness of technology-driven interventions in enhancing science learning among Grade 8 students at risk of dropping out in Alegria National High School, Philippines. The findings revealed that students who received technology-based instruction showed significantly higher gains in science knowledge compared to those who received traditional work-text interventions, highlighting the potential of technology in educational outreach (Pañares & Ybanez, 2021)

This research assessed the effects of technology integration on teaching performance and student engagement among public secondary school teachers in Bataan, Philippines. The study found that technology integration positively influenced teaching styles and student engagement, suggesting that a structured intervention framework could further enhance these outcomes (Quintos, 2024)

This research explores the preparedness of school leaders in adopting digital governance in Philippine educational institutions. Through surveys and interviews, the study identifies challenges such as limited resources and training, and proposes strategic interventions to enhance digital readiness, emphasizing the role of data-driven decision-making in educational leadership (Volante et al., 2025).

Theme 4: Technology Adoption in Public Education Monitoring Systems and Portals

This research details the development and implementation of a School Management Information System (SMIS) for a community college in Northern Mindanao. Employing the Agile Model and evaluated using the ISO 25010 quality model, the study demonstrates improvements in administrative efficiency, data management, and service delivery,

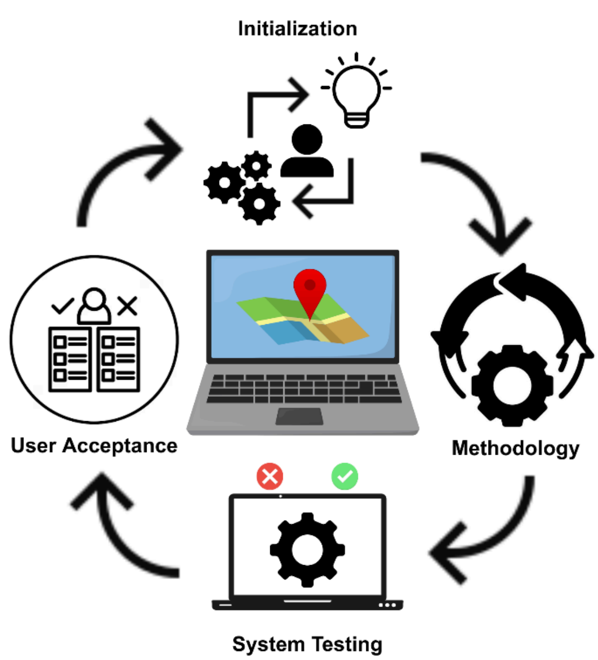
	highlighting the benefits of adopting centralized digital systems in educational institutions (Grepon et al., 2021). This systematic review examines the adoption of LMS platforms like Moodle and Google Classroom in Philippine secondary schools during the COVID-19 pandemic. The study highlights improvements in teaching performance and student achievement, emphasizing the role of LMS in facilitating distance learning and recommending institutional support for sustained technology integration (Borabo et al., 2024). This multiple-case study investigates the deployment of tablet computers in nine Philippine public schools under the Australian AID program. The research identifies school-level factors affecting technology use, including leadership, infrastructure, and training, offering recommendations for effective implementation of technology programs in developing-country contexts (Lumagbas et al., 2019). A student portal is a commonly used phrase to describe the login page where students can provide a username and password to gain access to an education organization's programs and other learning related materials. A student portal is essential in a digital-first education environment. As it centralizes all college-related announcements, information and resources. It also enables students to take care of all college-related needs from a single location, which means students have fewer accounts to manage. (Rupal et al., 2022) In accordance to the study of Uka (2019), the web-based student record management system for postsecondary institutions is the subject of this essay. The issues with student academic record management, such as incorrect course registration, delayed results release, result reconciliation, malpractice at various student clearing units, inaccuracy from laborious and manual calculations, and record retrieval challenges within the institution, led to the creation of this paper. This paper's goal is to provide a gateway that includes online profile creation and registration.
Conceptual Framework	

	Figure 1. Conceptual Framework of Tracking and Reaching ALS Learners System with Geo-mapping and Decision-Support System for Santa Cruz, Laguna
SWOT Analysis	

Strengths	**Weaknesses**	**Opportunities**	**Threats**
• Focuses on marginalized groups and aims to improve resource allocation and outreach. • Demonstrates a proactive approach to addressing existing challenges in ALS. • Directly addresses a real local issue in education accessibility in Sta. Cruz.	• False claims or inaccurate matches could result from the system's heavy reliance on user accuracy and honesty when submitting or claiming products. The lack of centralized admin verification can lead to The system's effectiveness hinges on the consistent cooperation of students, families, and field personnel in reporting and updating data. • Successful implementation and long-term maintenance require technical expertise and funding,	• The system can potentially be expanded to other regions if successful. • Encourages more evidence-based interventions in education planning.	• Collecting and mapping student data may raise concerns if not handled properly. • Stakeholders might resist shifting to a tech heavy system.

		which may not always be available at the local level.in conflict resolution or disagreements, and AI image recognition may also have trouble recognizing identical things if photographs are vague or generic.		

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